Student Final Marks Prediction

Predicting a student's final exam marks using machine learning

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Introduction

Schools record attendance, internal tests, assignments and study hours for every student, but rarely use them to guess the final exam marks. This project uses these five details to predict final marks out of 100.

Problem Statement

Teachers find out a student is weak only after the final exam, when it is too late to help.

The gap

No easy way to guess final marks early from the marks we already have.

The question

Can we predict final marks from attendance, two internal tests, assignment score and daily study hours?

Objectives

1. Study the student data and clean it.
2. Train four models to predict final marks.
3. Pick the model with the smallest error.
4. Build a simple app where anyone can enter marks and get a prediction.
5. Test that everything works correctly.

Scope of the Project

Coverage

Uses 5 inputs to predict 1 output (final marks). Includes data study, training, testing and a web app.

Target users

Teachers and mentors for an early estimate; students to check their own progress.

Limitations

Only about 2,000 records; error of 3-4 marks; shows the prediction, not the reason; no other details like subject or family.

Literature Review / Existing System

Existing approach	Limitation
Teacher's guess from marks	Not exact, takes time, gives no number
Average of test marks	Ignores attendance, assignment and study hours
Earlier ML models (Ridge, Random Forest, Gradient Boosting)	Often no easy app for teachers to use

Proposed System

Clean data steps

Fill missing values and put all numbers on the same scale.

Try 4 models

Linear, Ridge, Random Forest and Gradient Boosting tested on the same data.

Easy to use

Move the sliders and get predicted marks from 0 to 100.

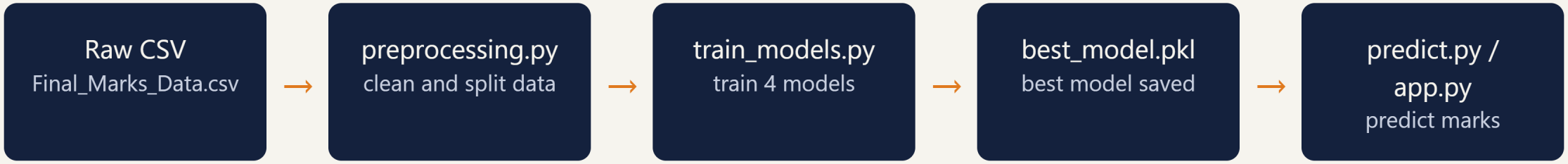
Shows what matters

Shows which factors affect marks the most.

Project Methodology

Stage	Tools
Language	Python
Data handling	pandas, NumPy; data kept in a CSV file (no database)
Models	scikit-learn (ready-made ML models)
Graphs	matplotlib, seaborn
Saving and app	joblib (save model), Streamlit (web app)
Testing	pytest

Project Internal Structure



Modules

config (settings), preprocessing (cleaning), eda (graphs), train_models, predict, app (web page)

Outputs

saved models, result tables, graphs, notebook

Implementation

- **preprocessing.py**: cleans the data
- **train_models.py**: trains models and picks the best
- **predict.py**: gives predicted marks (0-100)
- **app.py**: web page with sliders and a Predict button

Testing and Results

Model	Avg error	RMSE	Accuracy (R^2)	Re-check R^2
Ridge (best)	3.484	4.389	0.839	0.834
Linear Regression	3.484	4.389	0.839	0.834
Gradient Boosting	3.652	4.623	0.821	0.815
Random Forest	3.937	4.939	0.796	0.797

Conclusion

Our model predicts a student's final marks with an average error of about 3.5 marks, and it explains about 84% of how marks vary. All objectives are done: data studied, four models compared, the best one (Ridge) chosen, and a working app built and tested. Simple models worked as well as complex ones.

Future Scope

- Add more details like past results and class participation
- Collect more data from more classes
- Try improving and testing other models
- Warn students who are likely to fail
- Put the app online with a teacher login

References

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- Hoerl and Kennard, Ridge Regression, Technometrics 12(1), 1970
- Breiman, Random Forests, Machine Learning 45, 2001
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- Documentation: pandas, matplotlib, seaborn, joblib, Streamlit, pytest
- Dataset: Final_Marks_Data.csv [source- kaggle]